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UNIT-3

Database concepts & Structured Query Language (SQL)

1. What is Data?

Ans: Data is a collection/information of characters, numbers, and other symbols that represents values of some situations or variables

2.Types of Data

Ans: Data can be classified in to two types **(a) Structured data type**
(b) Unstructured data type

a) Structured data type: Data which is organised and can be recorded in a well defined format is called structured data. Structured data is usually stored in computer in a tabular (in rows and columns) format.

(b) Unstructured data type: Data, which are not in the traditional row and column Structure is called unstructured data.

Examples of unstructured data include web pages consisting of text as well as multimedia contents (image, graphics, and audio/video). Other examples include text documents, business reports, books, audio/video files, social media messages

3.Data Process Cycle:

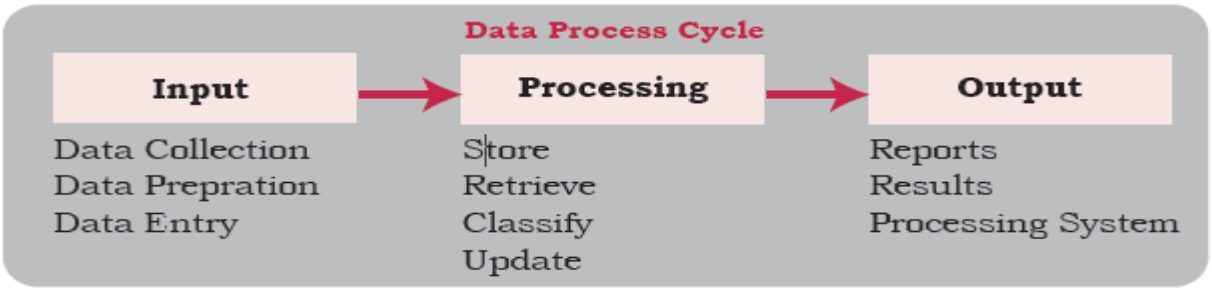


Figure 7.1: Steps in data processing

4.What is Manual record keeping or Database

Ans:A manual database is a paper-based database, where data is written on paper.

Examples of Manual Database

1) Entry of student details (Roll number and name) in the new attendance register when the student is promoted to the next class.
2) Writing student details on each month's attendance page where inconsistency may happen due to incorrectly written names, skipped student records, etc.
3) Loss of data in case attendance register is lost or damaged.
4) Erroneous calculation while consolidating attendance record manually.

5.What is File System

(a)A file can be understood as a container to store data in a computer. Files can be stored on the storage device of a computer system.
(b)File system becomes difficult to handle when number of files increases and volume of data also grows

Limitations/Drawbacks of a File System:

Following Six (6) are some of the limitations of file system:

1. Difficulty in access: Sometimes it is difficult to access data in the required format and one has to write application program to access data.
Ex: To find students who scored above 90%, every file must be searched separately.

Example:

Suppose a school maintains data in three files:

- `Students.txt` → contains Roll No, Name, Class
- `Marks.txt` → contains Roll No, Subject, Marks
- `Attendance.txt` → contains Roll No, Days Present

If the teacher wants to find **students of Class 11 who scored above 90% and had more than 90% attendance**, they must manually open all three files, match roll numbers, and calculate results — a time-consuming process.

2. Data Redundancy: Redundancy means same piece of data are **duplicated in different places** (files).Redundancy leads to excess storage use and may cause data inconsistency also
Ex: A student's address appears in both the student and fee record files.

Example:

In a school, data is stored in different text files:

- `Students.txt` → Roll No, Name, Class, Address
- `Marks.txt` → Roll No, Name, Subject, Marks
- `Attendance.txt` → Roll No, Name, Days Present

Here, the **Name** and **Roll No** of each student are repeated in every file.
If a student's name changes, it must be updated in all three files. If updated in one file but missed in others, it creates confusion.

Real-life Analogy:

It's like writing your address on multiple forms. If you move to a new house, you must update it everywhere — bank, school, ID cards, etc. If you forget one, old data remains in some places.

3. Data Inconsistency: Data inconsistency occurs when same data maintained in different places **do not match**. As the files are being maintained by different people, the changes may not happen in one of the file.
Ex: A student's new address is updated in one file but not in another.

Example:

A school keeps student data in different files:

- `Students.txt` → Roll No: 101, Name: *Rahul Verma*, Address: *Kurnool*
- `Marks.txt` → Roll No: 101, Name: *Rahul Varma*, Marks: 95

Here, the student's name is spelled differently in two files (*Verma* vs *Varma*). The system cannot identify which one is correct — this is **data inconsistency**.

4. Data Isolation: **Data isolation** means data is **stored in different files and formats**, making it difficult to combine or access together. Each file works **independently**, without a central link or connection between them..

Example:

A school stores information as follows:

- `Students.txt` → Roll No, Name, Class
- `Marks.csv` → Roll No, Subject, Marks
- `Fees.xlsx` → Roll No, Fees Paid

If the principal wants to find **students of Class 11 who paid all fees and scored above 90%**, it's difficult because the data is isolated in three separate files — text, CSV, and Excel — with no direct relationship.

+**Real-life Analogy:**

It's like keeping your clothes in one cupboard, shoes in another, and accessories in a third — when you need a complete outfit, you must search each place separately.

5.Data Dependence: Data are stored in a specific format or structure in a file. If the structure or format itself is changed, all the existing application programs accessing that file also need to be change. Otherwise, the programs may not work correctly. This is data dependency. Hence, updating the structure of a data file requires modification in all the application programs accessing that file.

Example:

Suppose a school has a file `Students.txt` with data stored as:

```
RollNo, Name, Class
```

If later the format changes to

```
RollNo, Name, Class, Address
```

then every Python or C++ program reading `Students.txt` must be edited to handle the new field. Otherwise, the program will show errors or wrong output.

6. Data Sharing: There can be different category of users to access particular data, ideally, **not every user should be able to access all the data**. It is very difficult to enforce this kind of access control in a file system while accessing files through application programs.

Example:

In a school:

- The **Accounts Department** keeps a file `Fees.xlsx`
- The **Examination Department** keeps a file `Marks.txt`
- The **Library** keeps a file `Students.csv`

If the principal wants to see all information about a student (fees, marks, and books issued), it cannot be easily shared between departments because each one maintains its own isolated file.

6. What is Database?

Ans: A database is a systematic collection of related data that is **stored, organized,** and **managed** so that it can be easily **accessed, retrieved**, and **updated** when needed.

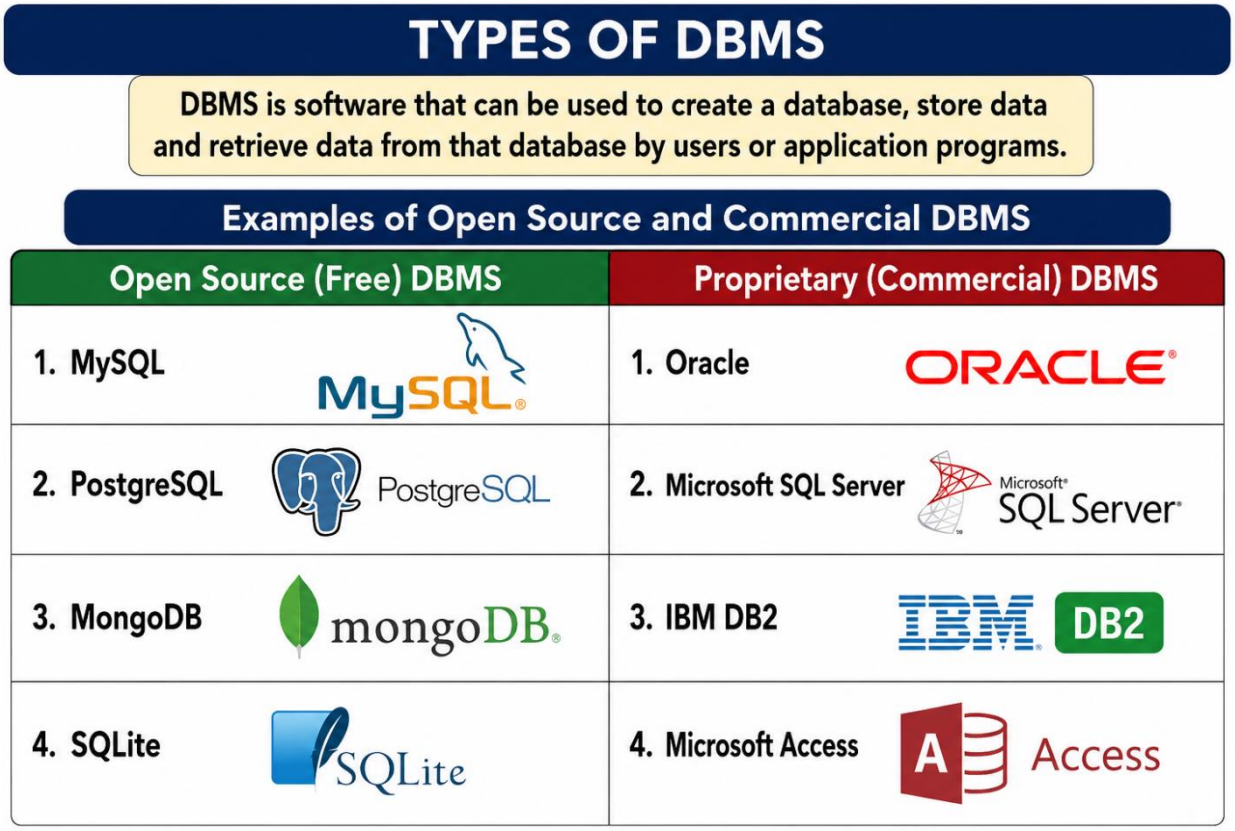
Or

Database is a collection of data which is stored in a systematic manner in the form of tables

7. What is DBMS (Database Management System)

DBMS is software that can be used to create a database, store, manage, update/modify and retrieve data from that database by users or application programs.

Some examples of open source and commercial DBMS include



8.What is RDBMS (Relational Database Management System)

RDBMS is software that executes queries on the related data, including adding, updating and searching for values.

9.Key Concepts in DBMS:In order to efficiently manage data using a DBMS, let us understand certain key terms

1.Data Constraint(Condition):is a **set of rules** that define valid data, Sometimes we put certain restrictions or limitations on the type of data that can be inserted in one or more columns of a table.

2. Data Dictionary*:* It is also called as **meta data** or **data about data**. The database schema along with various constraints on the data is stored by DBMS in a database catalogue.it helps data teams to understand data terms.

3. Database Engine: is a set of programs used by a DBMS to create database and handle various queries for data retrieval and manipulation.

4. Database Instance: is a snapshot of a database that exists at a particular time, the state of the database at any given time.

5.Data Manipulation(DML): Modification of database consists of three operations viz. Insertion, Deletion or Update.

6.Data Query(DQL): A query is a request to a database for obtaining information in a desired way

7. Database Schema (DESC): is the design/skeleton of a database represents the structure, type of column that holds constraints and relationship among tables.It is also called as visual or logical architecture.

7.4 Relational Data Model: Different types of DBMS are available and their classification is done based on the underlying data model

A data model describes the structure of the database, Including how data are defined and represented, relationships among data, and the constraints.

The most commonly used data model is **Relational Data Model**. The Relational data model was described in 1969 by computer scientist **Edgar .F.Codd**

Other types of data models include **Object-oriented data model, Entity-Relationship(E_R) data model**, **Document model** and **Hierarchical data model**.

In relational data model, tables are called relations that store data for different columns. Each table can have multiple columns where each column name should be unique.

A database that is modelled on relational data model concept is called Relational Database

Let us now understand the commonly used terminologies in relational data model

ATTRIBUTE: the columns of a relation (table) are the attributes, which are also referred as fields.

TUPLE: Each row of data in a relation (table) is called a Tuple/Record.

DOMAIN: A data type is used to specify domain for an attribute

DEGREE: The number of columns (attributes) in a relation is called the Degree

CARDINALITY: The number of tuples in a relation is called the Cardinality.

GENERAL	DBMS
Table	Relation
Row(Horizontal)	Tuple/Record
Column(Vertical)	Attributes/Fields
No.of Columns	Degree
No.of Rows	Cardinality
Structure	Schema /Design
Datatype	Domain

7.5 Keys in a Relational Database:

Q1) what is Key

Ans: A Key in DBMS is set of columns which helps you to identify a row in a table and helpful for finding record from the table and allows you to find the relation between two tables

Ex: Roll no, Aadhar card, Pan card, A/c No(Unique)

Q2) Types of Keys in Mysql

Ans: In MySQL there are **5** types of keys are there

a.Candidate key b.Primary Key c. Alternate Key

d. Composite Primary Key e. Foreign key

a. Candidate key: A candidate key is a **column** or a combination of columns uniquely identifying each row in a table.



Admno	Rno	Aadhar No	Name	Class
2025	1	123456781234	Raju	XI
2026	2	999988877777	Ramu	XI
2027	3	222233334444	Sita	XI
2028	4	555566667777	Joseph	XI
2029	5	888899990000	Rahim	XI

mysql> **CREATE TABLE student (Admno INT,Rno INT,AadharNo BIGINT, Name VARCHAR(50), Class VARCHAR(5));**

mysql>INSERT INTO student (Admno, Rno, AadharNo, Name, Class) VALUES
(2025, 1, 123456781234, 'Raju', 'XI'),
(2026, 2, 999988887777, 'Ramu', 'XI'),
(2027, 3, 222233334444, 'Sita', 'XI'),
(2028, 4, 555566667777, 'Joseph', 'XI'),
(2029, 5, 888899990000, 'Rahim', 'XI');

b. Primary Key: A primary key is a column that uniquely identifies each row in a table. No duplicates. No NULL.**(UNIQUE + NOT NULL)**

- The primary key should be selected from the candidate keys.
- Primary keys must contain unique values.
- A table **cannot** have more than one primary key.

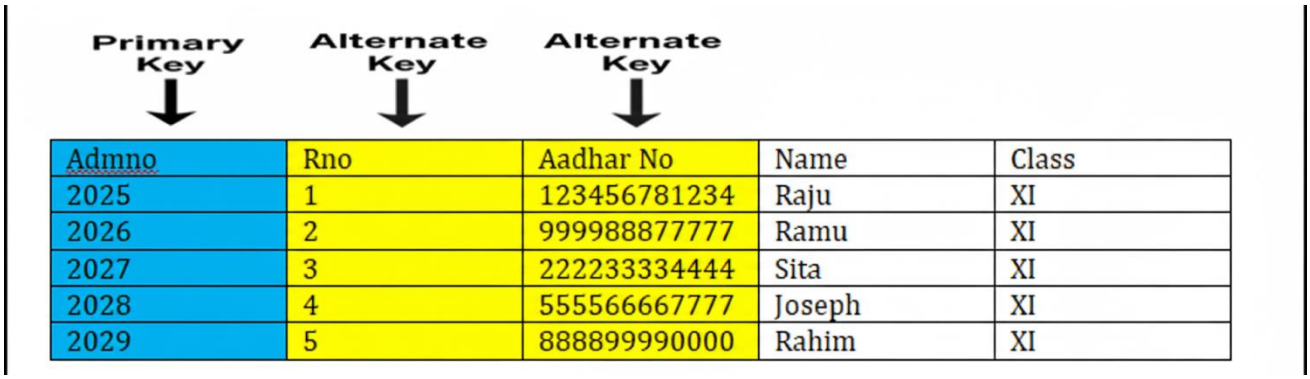


Admno	Rno	Aadhar No	Name	Class
2025	1	123456781234	Raju	XI
2026	2	999988877777	Ramu	XI
2027	3	222233334444	Sita	XI
2028	4	555566667777	Joseph	XI
2029	5	888899990000	Rahim	XI

mysql> CREATE TABLE student (Admno INT **PRIMARY KEY**,Rno INT,AadharNo INT, Name VARCHAR(50), Class VARCHAR(5));

mysql> INSERT INTO student (Admno, Rno, AadharNo, Name, Class) VALUES
(2025, 1, 123456781234, 'Raju', 'XI'),
(2026, 2, 999988877777, 'Ramu', 'XI'),
(2027, 3, 222233334444, 'Sita', 'XI'),
(2028, 4, 555566667777, 'Joseph', 'XI'),
(2029, 5, 888899990000, 'Rahim', 'XI');

C.Alternate Key: An Alternate Key is a Candidate Key that is ***not chosen*** **as** the Primary Key.



Admno	Rno	Aadhar No	Name	Class
2025	1	123456781234	Raju	XI
2026	2	999988877777	Ramu	XI
2027	3	222233334444	Sita	XI
2028	4	555566667777	Joseph	XI
2029	5	888899990000	Rahim	XI

d.Composite Primary Key: A composite key is a combination of **two or more** columns in a table that can be used to uniquely identify each row in the table.

Example to create a table using Composite Primary Key



Date	Rno	Status
2018-09-01	1	Present
2018-09-01	2	Present
2018-09-01	3	Absent
2018-09-02	1	Present
2018-09-02	2	Present
2018-09-03	3	Absent

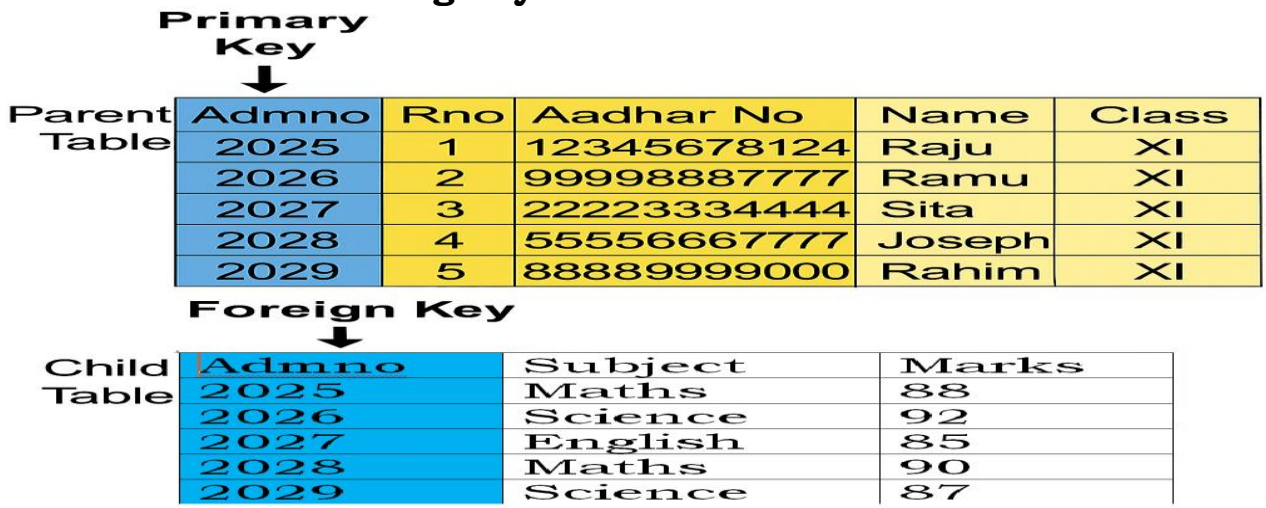
CREATE TABLE **attendance** (date DATE,rno INT,status VARCHAR(10), **PRIMARY KEY (date, rno)**);

INSERT INTO attendance (date, rno, status) VALUES
('2018-09-01', 1, 'Present'),('2018-09-01', 2, 'Present'),('2018-09-01', 3, 'Absent'),
('2018-09-02', 1, 'Present'),('2018-09-02', 2, 'Present'),('2018-09-03', 3, 'Absent');

e. Foreign Key: A foreign key is a column in one table (child table) that refers to the primary key of another table (parent table) to maintain referential integrity.

The table with the foreign key is called the **child table**, and the table with the primary key is called the referenced or **parent table**.

A foreign key is a key used to link **two** tables together. This is sometimes also called a **referencing key**.



Ex: **mysql>** CREATE TABLE student (Admno INT **PRIMARY KEY**,Rno INT,AadharNo INT, Name VARCHAR(50), Class VARCHAR(5));

mysql> INSERT INTO student (Admno, Rno, AadharNo, Name, Class) VALUES (2025, 1, 123456781234, 'Raju', 'XI'),(2026, 2, 999988877777, 'Ramu', 'XI'), (2027, 3, 222233334444, 'Sita', 'XI'),(2028, 4, 555566667777, 'Joseph', 'XI'), (2029, 5, 888899990000, 'Rahim', 'XI');

mysql>CREATE TABLE marks (Admno INT PRIMARY KEY,Subject VARCHAR(20),Marks INT,
FOREIGN KEY (Admno) REFERENCES student(Admno));

Inserting values in tables:
INSERT INTO marks (Admno, Subject, Marks) VALUES
(2025, 'Maths', 88),(2026, 'Science', 92),(2027, 'English', 85),
(2028, 'Maths', 90),(2029, 'Science', 87);

Foreign Key Errors
mysql> insert into marks values(2024,'Maths',90);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
mysql> insert into marks values(2025,'Maths',90);
ERROR 1062 (23000): Duplicate entry '2025' for key 'marks.PRIMARY'
mysql> DELETE FROM student WHERE Admno = 2025;
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails

Constraints in MySQL: Constraints are certain types of restrictions/conditions on the data values that an attribute can have

Constraint	Description
NOT NULL	Ensures that a column cannot have NULL values where NULL means missing/ unknown/not applicable value.
UNIQUE	Ensures that all the values in a column are distinct/unique.
DEFAULT	A default value specified for the column if no value is provided.
PRIMARY KEY	The column which can uniquely identify each row or record in a table.
FOREIGN KEY	The column which refers to value of an attribute defined as primary key in another table.

Not Null:CREATE TABLE **student_notnull** (Admno INT PRIMARY KEY, Name VARCHAR(50) **NOT NULL**,Class VARCHAR(10) **NOT NULL**);

mysql> insert into student_notnull values(1,null,'XI');
ERROR 1048 (23000): Column 'Name' cannot be null

UNIQUE:CREATE TABLE **student_with_unique** (AdmNo INT, Name VARCHAR(50), Class VARCHAR(10), Phone VARCHAR(15) **UNIQUE**);

DEFAULT:
CREATE TABLE **student_default** (AdmNo INT PRIMARY KEY, Name VARCHAR(50), Class VARCHAR(10) **DEFAULT 'XI'**, City VARCHAR(30) **DEFAULT 'Kurnool'**);

mysql> insert into student_default(Admno,Name) values(102,'Rafiq');

CHECK:
CREATE TABLE student_check (AdmNo INT PRIMARY KEY, Name VARCHAR(50), Age INT CHECK (Age >= 15), Marks INT CHECK (Marks BETWEEN 0 AND 100));

```
mysql> CREATE TABLE student (
    ->     AdmNo INT PRIMARY KEY,                  -- PRIMARY KEY
    ->     Name VARCHAR(50) NOT NULL,              -- NOT NULL
    ->     Email VARCHAR(50) UNIQUE,               -- UNIQUE
    ->     Class VARCHAR(10) DEFAULT 'XI',         -- DEFAULT
    ->     Age INT CHECK (Age >= 15)               -- CHECK
    -> );
```

Or

```
mysql> CREATE TABLE cons (
    ->     rno INT PRIMARY KEY,
    ->     phoneno VARCHAR(10) UNIQUE,
    ->     sname VARCHAR(15) NOT NULL,
    ->     gender CHAR(1) DEFAULT 'M' CHECK (gender IN ('M','F')),
    ->     marks INT CHECK (marks BETWEEN 0 AND 40)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

create table econs(eno int primary key,ename varchar(15),sal int check(sal between 10000 and 30000));

```
mysql> alter table dec18 add sal int check (sal>20000);
```

create table ex5(rno int,sname varchar(10), doa date check(doa between '2007-04-01' and '2023-03-30'));